SPORTS DATA CAMPUS

MSC IN ARTIFICIAL INTELLIGENCE APPLIED TO SPORTS

Module 6. Telling the game: NLP for sports intelligence

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UCAM UNIVERSIDAD CATÓLICA DE MURCIA

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Collaborative Assignment

Imagine that you have just joined an Artificial Intelligence team dedicated to real-time monitoring of the Olympic Games. As part of this team, your objective is to use Natural Language Processing (NLP) techniques to analyze a massive amount of textual data generated during the event: from social media posts to media reports and athlete interviews.

Your mission will be to transform these unstructured data into valuable information capable of providing insights into both the content itself and the emotions of the audience, as well as the narratives that emerge throughout the competition.

This challenge will allow you to apply your NLP knowledge in a real and dynamic environment, contributing to the advanced analysis of one of the most important sporting events in the world.

Objective

Each student must choose one topic related to the Olympic Games, search for content in articles, social media, news, or other sources, and apply NLP techniques to process and analyze the data.

At the end, students must share their results with their classmates, encouraging an exchange of ideas and experiences.

Assignment Instructions

1. Choose one specific topic related to the Olympic Games:

(1 single topic)
Examples of topics:

- Fan sentiment on social media regarding an athlete's performance
- Comparison of articles from different media outlets about a specific Olympic event
- Analysis of athlete interviews and their speeches after victories or defeats

2. Data Research:

(1 single data source)
Each student must search for relevant content sources, such as:

- Social media: Posts on Twitter, Instagram, or Facebook about the Olympic Games
- Articles: Sports reports, interviews, or analyses about events and athletes
- News: Press coverage from past or current editions of the Games
- Other sources: Blogs, sports forums, or sports databases

3. NLP Techniques to Apply:

(select 2 techniques)

- Tokenization and Text Preprocessing: Clean and prepare the data for analysis
- Sentiment Analysis: Assess the general perception of an athlete, country, or event
- Named Entity Recognition (NER): Identify athletes, countries, competitions, and locations
- Word Frequency Analysis: Count how often certain words appear in a text
- Keyword Extraction: Identify the most important words in a text

4. Project Development

- Apply 2 NLP techniques that you consider the most relevant to your topic
- Document every step of the process: data collection, preprocessing, applied technique, and obtained results
- Include visualizations (charts, word clouds, etc.) that help illustrate your findings

5. Sharing

At the end, present your results to the group, explaining:

- How you found the data and preprocessed it
- Which NLP techniques you used and why
- What results and insights you obtained from the analysis

Compare your results with those of your classmates and collaborate to enrich the discussion.